

2026-08-02 · 56 features · fitted on 198,157 human decisions from 7,507 clean open-sheet games

GENERATED by `build/build_mag_variables.js` from the live weight file. Do not hand-edit the tables.

Nobody typed these numbers. Each is fitted: the model is shown a real human decision and all the options that person had, and the weights are nudged until the option they actually clicked comes out most likely. So every weight answers one question — **what makes a human more likely to click this** — and never *what wins*.

`ZERO` marks a weight whose 95% interval contains zero: the data cannot tell it from no effect.

The last column is the same fit with **popularity removed entirely**.

On held-out human decisions the full model picks the option the person actually clicked **32.2%** of the time; without popularity, **31.3%** — **a loss** of 1.0 points. So popularity carries real signal and is not crowding the board facts out. An earlier version of this document asserted the opposite, from a fit whose `DROP=` reached only half the corpus.

feature	weight	no-popularity	what it means
<code>eff4</code>	+1.177	+1.127	it hits a 4x weakness
<code>eff2</code>	+0.859	+0.830	it hits a 2x weakness
<code>stab</code>	+0.055	+0.208	it matches my own type
<code>allyHit</code>	-0.082	-0.139	it also hits my own partner, and my partner is not immune to it
<code>bp</code>	-0.170	-0.128	it is a powerful move
<code>effHalf</code>	-0.745	-0.741	it is resisted
<code>accuracy</code>	-0.795	-0.874	how often it hits, on THIS board (snow makes Blizzard certain)
<code>effQuarter</code>	-0.885	-0.920	it is resisted twice over
<code>immune</code>	-1.928	-1.899	it does nothing at all
<code>abilityBlock</code>	-2.077	-2.062	the target probably has an ability that eats it

feature	weight	no-popularity	what it means
<code>rechargeTurn</code>	-0.393	-0.475	it costs me the turn AFTER this one
<code>chargeTurn</code>	-0.902	-0.838	it needs a turn to wind up on this board

feature	weight	no-popularity	what it means
<code>isStatus</code>	+0.260	+0.278	it is a status move
<code>tgtHurt</code>	+0.097	+0.088	the target is already hurt
<code>pivots</code>	-0.178	-0.067	it damages and brings me out
<code>deadWeather</code>	-1.001	-1.097	that weather is already set, so it would fail
<code>deadStall</code>	-1.292	-1.273	I protected last turn, so it would probably fail
<code>deadStatus</code>	-1.607	-1.489	the target already has a status, so it would fail
<code>deadField</code>	-1.662	-1.286	that field effect is already up, so it would fail
<code>pranksterFailsDark</code>	-1.992	-1.903	my Prankster status move is aimed at a Dark type, so it does nothing
<code>deadNoLastMove</code>	-2.673	-2.603	it needs the target to have already moved, and the target just switched in
<code>deadSide</code>	-2.960	-2.884	that side effect is already up, so it would fail

feature	weight	no-popularity	what it means
<code>defMismatch</code>	+0.162	+0.161	I hit its softer defence
<code>tgtBulk</code>	+0.124	+0.139	the target is bulky
<code>movesFirst</code>	+0.040	+0.055	I move before the target, counting priority, Tailwind and Trick Room
<code>priority</code>	-0.075	+0.080	this move cuts the queue (Fake Out, Sucker Punch, Extreme Speed)

feature	weight	no-popularity	what it means
tgtPhysical	-0.189	-0.180	the target attacks physically rather than specially (negative = specially)

feature	weight	no-popularity	what it means
killIsRoll	+0.783	+0.845	it kills some spreads and not others: a roll rather than a read
tgtMayProtect	+0.408	+0.440	how often this target blocks: the biggest reason a sure kill is not one
protectThreatened	+0.320	+0.317	this move protects, and I am facing a kill
dmgFrac	+0.274	+0.281	how much of what is left of the target it takes
koFirst	+0.143	+0.145	it kills, and I move first, so the kill lands before their attack
koTarget	+0.014 ZERO	+0.031	the odds this really kills it: the worst roll still does, and the move lands
killsThreat	-0.057	-0.058	it kills the thing that was about to kill me

feature	weight	no-popularity	what it means
volatileOnSelf	+0.123	+0.138	it puts something on me or my side (Substitute, Follow Me, Rage Powder)
volatileOnFoe	-0.151	-0.173	it takes an option away from the target (Taunt, Encore, Disable)

feature	weight	no-popularity	what it means
myOffenseStage	+1.201	+1.214	how boosted the stat I am attacking with already is
movesBoostMe	+0.609	+0.691	this move raises one of my own stats
tgtDefenseStage	+0.132 ZERO	+0.161	how boosted the defence I am attacking into already is
movesLowerFoe	-0.316	-0.696	this move lowers one of the target's stats

feature	weight	no-popularity	what it means
statusBites	+1.084	+0.775	the status this inflicts hits a stat the target actually relies on

feature	weight	no-popularity	what it means
diesBeforeMoving	-0.582	-0.590	I am facing a kill and I do not move first

feature	weight	no-popularity	what it means
switchSurvives1	+0.193	+0.193	the replacement lives through the hardest thing aimed at me
switchSurvives2	-0.004 ZERO	-0.009	it lives through that twice
switchFaster	-0.202	-0.194	the replacement outruns the thing that was threatening me
isSwitch	-0.275	-0.984	this candidate is a switch, not a move

feature	weight	no-popularity	what it means
switchKOSlow	+0.102	+0.074	it survives what it must eat, moves second, and still removes a foe
switchKOFast	-0.017 ZERO	-0.047	it survives what it must eat, moves FIRST, and removes a foe
switchDiesFirst	-0.133	-0.152	it does NOT survive what it must eat: it never gets to act at all

feature	weight	no-popularity	what it means
healValue	+2.428	+2.399	it heals me AND I am hurt enough for the healing not to be wasted
screenValue	+1.156	+1.231	it halves incoming damage AND something across from me is actually hitting
speedSwing	+0.958	+1.163	this move flips the speed order IN MY FAVOUR
priorLogP	+0.154	+0.000	it is a popular move
benchRisk	-0.155	-0.138	1 - worst-case value retention if their bench answers this click
clickCost	-0.248	-0.236	expected self-cost of this click into that body (fraction-of-self units)
stallIntoEncore	-0.848	-0.823	I am about to Protect and something across from me can Encore me for it